

Rohith Reddy Kolanu

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Summary

Analytics professional with **2 years** of business intelligence and reporting automation experience at Broadridge, recently completed graduate study in Business Analytics and AI/ML. Skilled in Python, SQL, Power BI, machine learning, neural networks, & NLP through applied projects in predictive modeling. Seeking a data science role that bridges business and data.

Education

The University of Texas at Austin

Austin, TX

Post Graduate Program in Artificial Intelligence and Machine Learning, CGPA: 4.33/4.33

Dec 2025 - Aug 2026

Relevant Coursework: Python Foundations, Machine Learning, Advanced Machine Learning, Introduction to Neural Networks, Natural Language Processing with Generative AI, Introduction to Computer Vision, Model Deployment

University of Massachusetts Amherst

Amherst, MA

Master of Science in Business Analytics, Data Analytics, CGPA: 4.0/4.0

Aug 2024 - May 2026

Relevant Coursework: Data Management, Business Intelligence, Business Applications with Python, Machine Learning, Project Management, Financial Technology, Data Science, Data Visualization, Artificial Intelligence

Experience

Broadridge

Hyderabad, India

Process Analyst

Aug 2022 - Jul 2024

- Designed and fully automated an Excel-based utilization tracking system that standardized calculations, cut manual effort by **90 percent**, and fed workforce analytics into monthly Power BI and PowerPoint reports.
- Built and delivered monthly performance dashboards in Power BI and Excel that improved reporting accuracy and visibility into the key utilization metrics, trends, and KPIs for data-driven operational decisions.
- Analyzed and processed over **2,000 client documents**, websites, and emails at **99% accuracy**, ensuring SLA compliance while maintaining strong data quality, validation standards, and end-to-end process consistency.
- Handled **300+ operational requests** during peak cycles at **99% accuracy**, trimming turnaround time by **20%** through workflow optimization, prioritization logic, and continuous process improvements across every cycle.
- Collaborated cross-functionally with the Desktop Publishing and Client-Facing teams to validate data outputs, strengthen reporting reliability, and turn operational data into clear actionable business insights.

Projects

MedRAG: Retrieval Augmented Clinical Decision Support Powered by the Merck Manuals | [GitHub](#)

- Built an end to end clinical RAG pipeline over the **4,114 page** Merck Manual, chunking it into **8,760 segments** with **384 dimensional embeddings** in ChromaDB and serving grounded answers through a locally hosted Mistral 7B model.
- Engineered retrieval and prompt workflows over **8,760 chunks** with similarity search, context injection, and instruction formatting, then scored answers on groundedness and relevance, cutting hallucination on high risk drug queries.

Airbnb Price Prediction, Guest Sentiment Analysis & Topic Modeling using ML & NLP | [GitHub](#)

- Built an end-to-end Airbnb price prediction pipeline on **3,840 Chicago listings** using structural and NLP features, trained and tested **2 neural networks** and tuned XGBoost achieving **0.61 test R-squared** and **\$81.2 RMSE** as the top model.
- Engineered an NLP pipeline on **271,871 reviews** with VADER, DistilBERT sentiment, zero-shot scoring across **five categories**, and LDA, NMF, LSA, BERTopic modeling, turning unstructured feedback into listing-level pricing signals.

Predicting Visa Approval Outcomes Using Advanced Machine Learning Techniques | [GitHub](#)

- Built an end-to-end visa approval prediction pipeline on **25,480 records** using applicant and employer features, tested **5 ensemble classifiers** with tuned XGBoost achieving **74.25% accuracy** and **0.819 F1-score** as the top performer.
- Engineered a preprocessing pipeline with one-hot encoding, SMOTE-based class balancing, stratified splitting, and model tuning, **optimizing recall to 87.31%** to capture certified applications and streamline manual reviews.

Technical Skills

Programming Languages & Databases: Python, R, SQL, MSSQL, PostgreSQL, MySQL, NoSQL, Git

Data Analysis & Visualization: Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly Express, Excel, Power BI, Tableau

Machine Learning & AI: Scikit-learn, Regression, Classification, Clustering, Deep Learning, Neural Networks, NLP

Deep Learning & GenAI: TensorFlow, Keras, PyTorch, LLMs, Hugging Face, LangChain, ChromaDB

Big Data & Cloud Platforms: Apache Spark, Kafka, Airflow, Hadoop, ETL/ELT Pipelines, Snowflake, AWS

Model Deployment & MLOps: Docker, Flask, Google Colab, Jupyter Notebook, OpenCV, spaCy